INCH-POUND

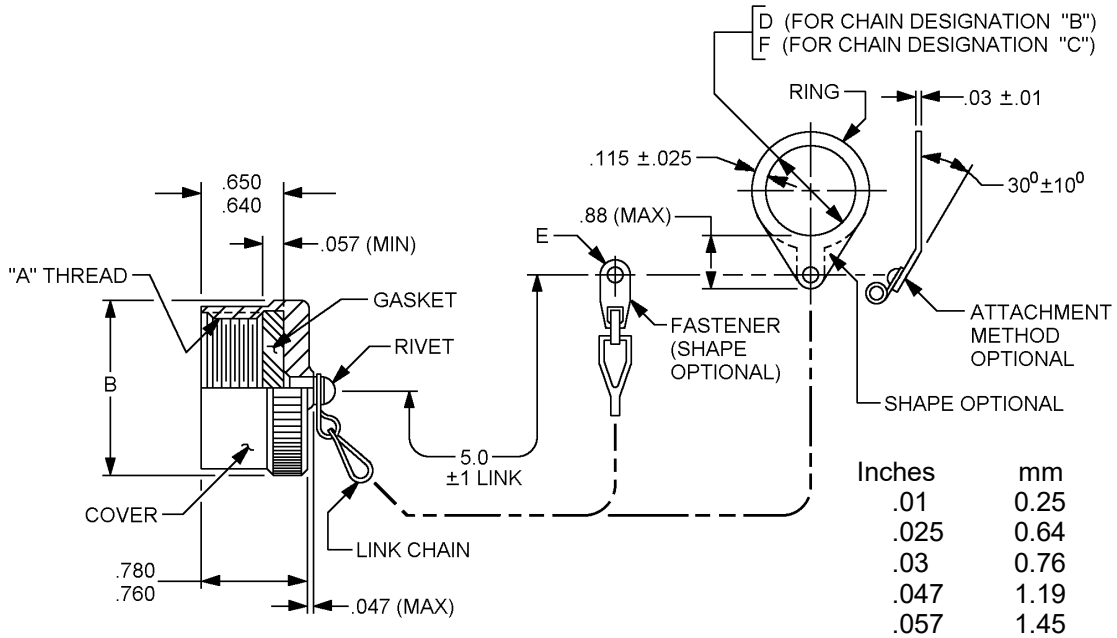
MIL-DTL-28840/13F
w/AMENDMENT 2
03 March 2021
SUPERSEDING
MIL-DTL-28840/13F
w/AMENDMENT 1
2 May 2019

DETAIL SPECIFICATION SHEET

CONNECTORS, ACCESSORIES, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, PROTECTIVE COVER, RECEPTACLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.



Inches	mm
.01	0.25
.025	0.64
.03	0.76
.047	1.19
.057	1.45
.115	2.92
.640	16.26
.650	16.51
.760	19.30
.780	19.81
.88	22.35
5.00	127.00

FIGURE 1. Dimensions and configurations.

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Shell size designator (see note 1)	A thread class 2B	B max dia	D ± .010 (± .25)	E ± .005 (± .13)	F ± .005 (± .13)
A (11)	.750-1P-.2L-DS	1.028 (26.11)	.890 (22.61)	.130 (3.30)	.765 (19.43)
B (13)	.875-1P-.2L-DS	1.141 (28.98)	1.015 (25.78)		.890 (22.61)
C (15)	1.062-1P-.2L-DS	1.263 (32.08)	1.203 (30.56)		1.015 (25.78)
D (17)	1.125-1P-.2L-DS	1.387 (35.23)	1.265 (32.13)		1.140 (28.96)
E (19)	1.312-1P-.2L-DS	1.513 (38.43)	1.453 (36.91)		1.265 (32.13)
F (23)	1.500-1P-.2L-DS	1.703 (43.26)	1.640 (41.66)		1.453 (36.91)
G (25)	1.625-1P-.2L-DS	1.825 (46.35)	1.765 (44.83)	.156 (3.96)	1.577 (40.06)
H (29)	1.812-1P-.2L-DS	2.143 (54.43)	1.953 (49.61)		1.890 (48.01)
J (33)	2.000-1P-.2L-DS	2.329 (59.16)	2.140 (54.36)	.193 (4.90)	2.077 (52.76)

NOTES:
1. Shell sizes are provided within parentheses for information and are not a part of the designator.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Dimensions apply after plating.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Interface dimensions shall conform to MIL-DTL-28840.

Chain shall be free to rotate on rivet. Chain material shall be 300 series corrosion resistant steel, passivated in accordance with SAE-AMS2700.

Gasket shall be mechanically or chemically captive to the cover. Gasket material shall be fluorosilicone.

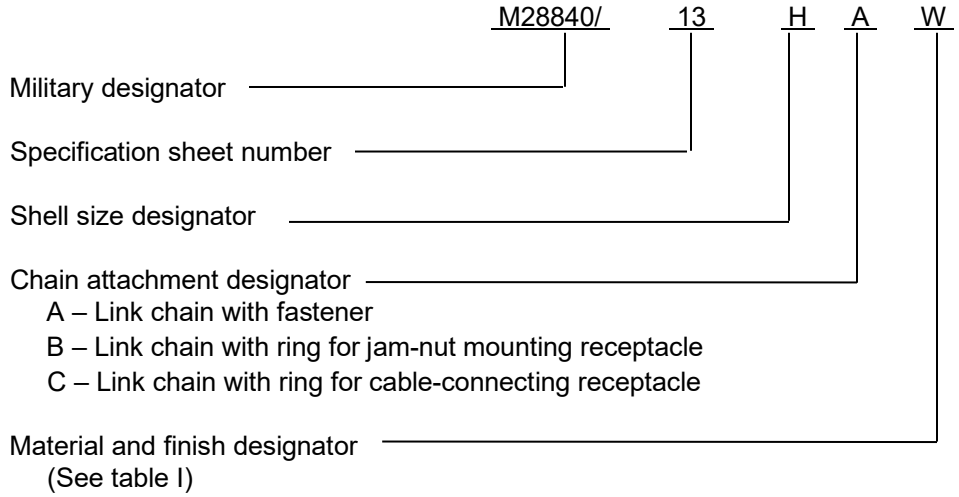
Material and finish, protective covers: See table I.

TABLE I. Material and finish.

Material and finish designator	Material	Finish
W	Aluminum alloy in accordance with ASTM B85/B85M, ASTM B26/B26M, ASTM B108/B108M, ASTM B209, ASTM B211 or ASTM B221	Cadmium plate, olive drab over electroless nickel
D	Corrosion resistant steel in accordance with ASTM A276, ASTM A473, or ASTM A580/A580M, type 316	Cadmium plate, black
F	Aluminum alloy in accordance with ASTM B85/B85M, ASTM B26/B26M, ASTM B108/B108M, ASTM B209, ASTM B211 or ASTM B221	Pure electro-deposited aluminum, non-reflective
L		Tin-zinc, non-reflective
S		Zinc nickel, non-reflective

Marking: The cover shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

PIN example:



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QUALIFICATION: One protective cover of each shell size for which qualification is desired shall have the following tests performed in the sequence shown:

Visual and mechanical.

Magnetic permeability.

Shell conductivity.

Corrosion.

Air leakage: Cover shall be mated to suitable connector plug and tested in accordance with test procedure EIA-364-02. The following details shall apply: The air leakage rate shall not be greater than one atmospheric cubic inch per hour (4.55×10^{-3} cubic centimeter per second).

Tensile strength: Cover shall be mounted to a suitable fixture to facilitate a 25-pound pull applied at the rate of 5 pounds per second and held for a period of 5 seconds. The load shall be applied to the chain end fastener through the center line axis of the protective cover, and then repeated through a perpendicular axis (90°) from the center line axis. Chain or fastener breakage, or loosening of the rivet, shall be considered failures.

Shell conductivity.

Post visual and mechanical.

RESTRICTIONS:

Material and finish designators W and D are inactive for NAVSEA new design, unless approved by NAVSEA.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-28840, this document references the following:

ASTM A276
ASTM A473
ASTM A580/A580M
ASTM B26/B26M
ASTM B85/B85M
ASTM B108/B108M
ASTM B209
ASTM B211
ASTM B221
EIA-364-02
SAE-AMS2700

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2021-013)

Review activities:
Army – AV, MI
Navy - OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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